5 .Hadoop Map Reduce

Input File : TCC48.txt

Contents:

```
HDFS is a storage unit of Hadoop
MapReduce is a processing tool of Hadoop
```

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ROLL NO : TCC48

```
WC_Mapper.java

package com.javatpoint;
import java.io.IOException;
import java.util.StringTokenizer;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;
public class WC_Mapper extends MapReduceBase implements Mappe
r<LongWritable,Text,Text,IntWritable>{
private final static IntWritable one = new IntWritable(1);
private Text word = new Text();
public void map(LongWritable key, Text value,OutputCollector<
Text,IntWritable>
output,
Reporter reporter) throws IOException{
String line = value.toString();
StringTokenizer tokenizer = new StringTokenizer(line);
```

```
while (tokenizer.hasMoreTokens()){
word.set(tokenizer.nextToken());
output.collect(word, one);
}
} }
```

WC_Reducer.java

```
package com.javatpoint;
import java.io.IOException;
import java.util.Iterator;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;
public class WC_Reducer extends MapReduceBase implements
Reducer<Text,IntWritable,Text,IntWritable> {
public void reduce(Text key, Iterator<IntWritable>
values,OutputCollector<Text,IntWritable> output,
Reporter reporter) throws IOException {
int sum=0;
while (values.hasNext()) {
sum+=values.next().get();
}
output.collect(key,new IntWritable(sum));
} }
```

WC_Runner.java

```
package com.javatpoint;
import java.io.IOException;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.mapred.TextInputFormat;
import org.apache.hadoop.mapred.TextOutputFormat;
public class WC_Runner {public static void main(String[] arg
s) throws IOException{
                JobConf conf = new JobConf(WC_Runner.class);
                conf.setJobName("WordCount");
                conf.setOutputKeyClass(Text.class);
                conf.setOutputValueClass(IntWritable.class);
                conf.setMapperClass(WC_Mapper.class);
                conf.setCombinerClass(WC_Reducer.class);
                conf.setReducerClass(WC_Reducer.class);
                conf.setInputFormat(TextInputFormat.class);
                conf.setOutputFormat(TextOutputFormat.class);
                FileInputFormat.setInputPaths(conf,new Path(a
rgs[0]));
                FileOutputFormat.setOutputPath(conf,new Path
(args[1]));
                JobClient.runJob(conf);
            }
        }
```

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hadoop@student: ~/hadoop

```
student@student:~$ su - hadoop
Password:
hadoop@student:~$ cd hadoop
hadoop@student:~/hadoop$ hadoop version
Hadoop 3.3.6
Source code repository https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195058f08fd012bf9c
Compiled by ubuntu on 2023-06-18T08:22Z
Compiled on platform linux-x86_64
Compiled with protoc 3.7.1
From source with checksum 5652179ad55f76cb287d9c633bb53bbd
This command was run using /home/hadoop/hadoop/share/hadoop/common/hadoop-common-3.3.6.jar
hadoop@student:~/hadoop$ nano TCC48.txt
hadoop@student:~/hadoop$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [student]
Starting resourcemanager
Starting nodemanagers
hadoop@student:~/hadoop$ hdfs dfs -mkdir /test_wc
hadoop@student:~/hadoop$ hdfs dfs -put admin:///home/hadoop/hadoop/TCC51.txt/test_wc
put: `.': No such file or directory: `hdfs://localhost:9000/user/hadoop'
hadoop@student:~/hadoop$ ls
bin       lib             licenses-binary  NOTICE-binary  sbin
etc       libexec         LICENSE.txt      NOTICE.txt     share
include   LICENSE-binary  logs             README.txt     TCC51.txt
hadoop@student:~/hadoop$ hdfs dfs -put admin:///home/hadoop/hadoop/TCC51/test_wc
put: `.': No such file or directory: `hdfs://localhost:9000/user/hadoop'
hadoop@student:~/hadoop$ hdfs dfs -put admin:///home/hadoop/hadoop/TCC51.txt /test_wc
hadoop@student:~/hadoop$ Ifconfig
Command 'Ifconfig' not found, did you mean:
  command 'fconfig' from deb redboot-tools (0.7build4)
  command 'ifconfig' from deb net-tools (1.60+git20181103.0eebece-1ubuntu5.4)
Try: apt install <deb name>
hadoop@student:~/hadoop$ Ifconfig
Command 'Ifconfig' not found, did you mean:
```

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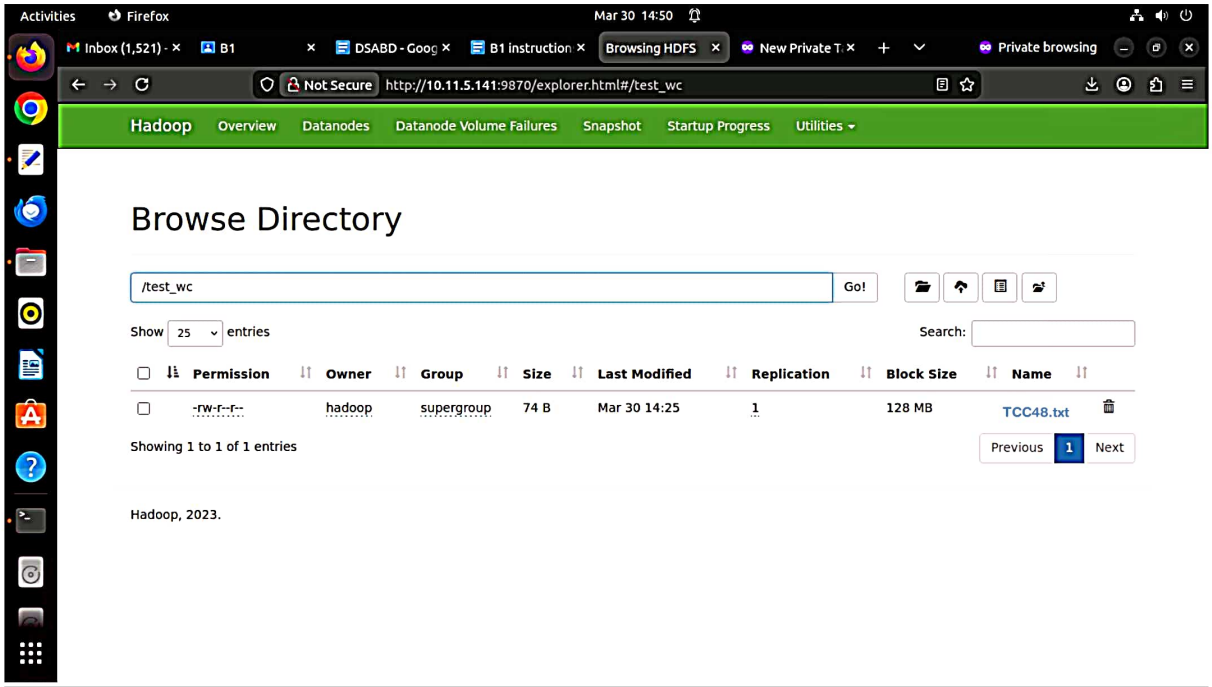
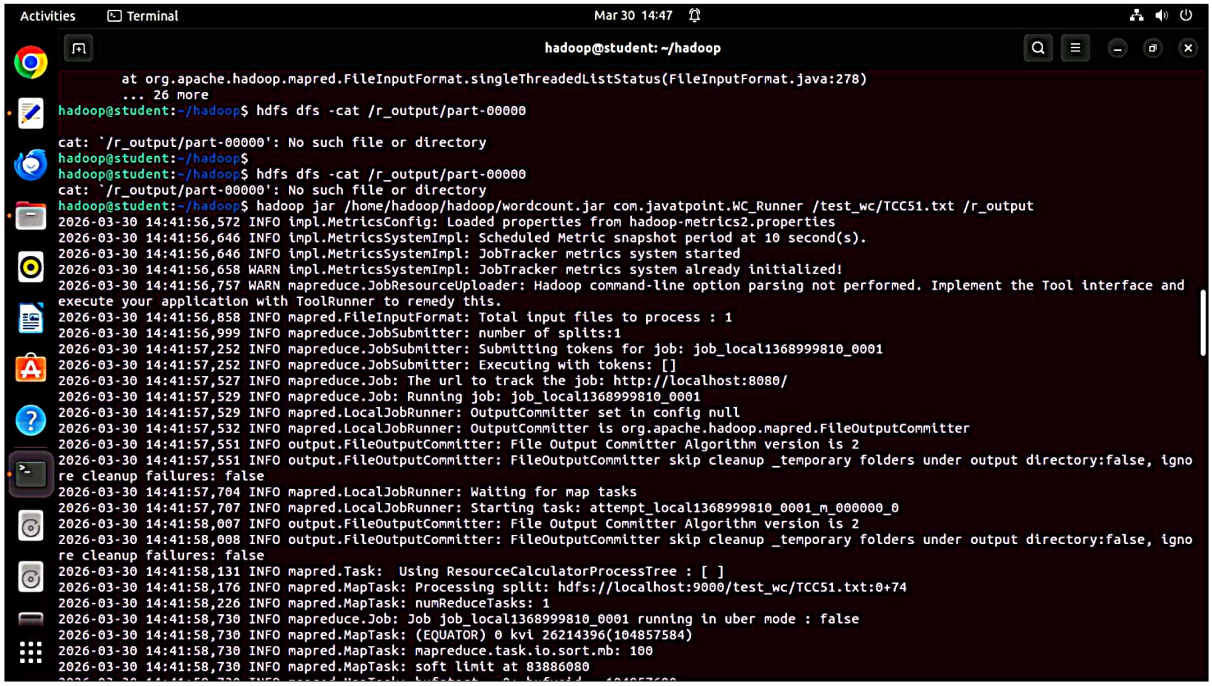
hadoop@student: ~/hadoop

```
Try: apt install <deb name>
hadoop@student:~/hadoop$ ifconfig
eno1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.11.5.141  netmask 255.255.0.0  broadcast 10.11.255.255
        inet6 fe80::9776:a8e7:63a3:89ef  prefixlen 64  scopeid 0x20<link>
        ether 50:65:f3:32:cd:0d  txqueuelen 1000  (Ethernet)
        RX packets 59816  bytes 52239741 (52.2 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 34657  bytes 9383450 (9.3 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
        device interrupt 20  memory 0xf7c00000-f7c20000

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 3592  bytes 462298 (462.2 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 3592  bytes 462298 (462.2 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

hadoop@student:~/hadoop$ ^C
hadoop@student:~/hadoop$ nano WC_Mapper.java
hadoop@student:~/hadoop$ nano WC_Reducer.java
hadoop@student:~/hadoop$ Nano WC_Runner.java
Command 'Nano' not found, did you mean:
  command 'nano' from snap nano (7.2+pkg-4057)
  command 'nano' from deb nano (6.2-1ubuntu0.1)
See 'snap info <snapname>' for additional versions.
hadoop@student:~/hadoop$ nano WC_Runner.java
hadoop@student:~/hadoop$ javac -classpath "$(hadoop classpath)" -d . WC_Mapper.java WC_Reducer.java WC_Runner.java
hadoop@student:~/hadoop$ jar -cvf wordcount.jar com
added manifest
adding: com/(in = 0) (out= 0)(stored 0%)
adding: com/javatpoint/(in = 0) (out= 0)(stored 0%)
adding: com/javatpoint/WC_Mapper.class(in = 1874) (out= 768)(deflated 59%)
adding: com/javatpoint/WC_Reducer.class(in = 1546) (out= 613)(deflated 60%)
adding: com/javatpoint/WC_Runner.class(in = 1485) (out= 728)(deflated 50%)
```


Activities
Firefox
Mar 30 14:50
Inbox (1,521) - ×
B1
DSABD - Goog ×
B1 instruction ×
Browsing HDFS ×
New Private T ×
Private browsing
Not Secure
http://10.11.5.141:9870/explorer.html#/test_wc
Hadoop
Overview
Datanodes
Datanode Volume Failures
Snapshot
Startup Progress
Utilities
Browse Directory
/test_wc
Go!
Show 25 entries
Search:
Permission
Owner
Group
Size
Last Modified
Replication
Block Size
Name
-rw-r--r--
hadoop
supergroup
74 B
Mar 30 14:25
1
128 MB
TCC48.txt
Showing 1 to 1 of 1 entries
Previous
1
Next
Hadoop, 2023.


Activities
Terminal
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hadoop@student: ~/hadoop
at org.apache.hadoop.mapred.FileInputFormat.singleThreadedListStatus(FileInputFormat.java:278)
... 26 more
hadoop@student:~/hadoop$ hdfs dfs -cat /r_output/part-00000
cat: `/r_output/part-00000': No such file or directory
hadoop@student:~/hadoop$
hadoop@student:~/hadoop$ hdfs dfs -cat /r_output/part-00000
cat: `/r_output/part-00000': No such file or directory
hadoop@student:~/hadoop$ hadoop jar /home/hadoop/hadoop/wordcount.jar com.javatpoint.WC_Runner /test_wc/TCC51.txt /r_output
2026-03-30 14:41:56,572 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-03-30 14:41:56,646 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-03-30 14:41:56,646 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-03-30 14:41:56,658 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-03-30 14:41:56,757 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-03-30 14:41:56,858 INFO mapred.FileInputFormat: Total input files to process : 1
2026-03-30 14:41:56,999 INFO mapreduce.JobSubmitter: number of splits:1
2026-03-30 14:41:57,252 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1368999810_0001
2026-03-30 14:41:57,252 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-30 14:41:57,527 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-03-30 14:41:57,529 INFO mapreduce.Job: Running job: job_local1368999810_0001
2026-03-30 14:41:57,529 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-03-30 14:41:57,532 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapred.FileOutputCommitter
2026-03-30 14:41:57,551 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-30 14:41:57,551 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-03-30 14:41:57,704 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-03-30 14:41:57,707 INFO mapred.LocalJobRunner: Starting task: attempt_local1368999810_0001_m_000000_0
2026-03-30 14:41:58,007 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-30 14:41:58,008 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-03-30 14:41:58,131 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-03-30 14:41:58,176 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/test_wc/TCC51.txt:0+74
2026-03-30 14:41:58,226 INFO mapred.MapTask: numReduceTasks: 1
2026-03-30 14:41:58,730 INFO mapreduce.Job: Job job_local1368999810_0001 running in uber mode : false
2026-03-30 14:41:58,730 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-03-30 14:41:58,730 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-03-30 14:41:58,730 INFO mapred.MapTask: soft limit at 83886080

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Screenshot captured
You can paste the image from the clipbord.

```
2026-03-30 14:41:59,938 INFO mapred.MapTask:
2026-03-30 14:41:59,938 INFO mapred.MapTask:
2026-03-30 14:41:59,938 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214344(104857376); length = 53/6553600
2026-03-30 14:41:59,966 INFO mapred.MapTask: Finished spill 0
2026-03-30 14:42:00,014 INFO mapred.Task: Task:attempt_local1368999810_0001_m_000000_0 is done. And is in the process of committing
2026-03-30 14:42:00,019 INFO mapred.LocalJobRunner: hdfs://localhost:9000/test_wc/TCC51.txt:0+74
2026-03-30 14:42:00,019 INFO mapred.Task: Task 'attempt_local1368999810_0001_m_000000_0' done.
2026-03-30 14:42:00,026 INFO mapred.Task: Final Counters for attempt_local1368999810_0001_m_000000_0: Counters: 24
        File System Counters
                FILE: Number of bytes read=3228
                FILE: Number of bytes written=645188
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=74
                HDFS: Number of bytes written=0
                HDFS: Number of read operations=5
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=1
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=2
                Map output records=14
                Map output bytes=130
                Map output materialized bytes=125
                Input split bytes=91
                Combine input records=14
                Combine output records=10
                Spilled Records=10
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=12
                Total committed heap usage (bytes)=227540992
        File Input Format Counters
                Bytes Read=74
2026-03-30 14:42:00,026 INFO mapred.LocalJobRunner: Finishing task: attempt_local1368999810_0001_m_000000_0
2026-03-30 14:42:00,027 INFO mapred.LocalJobRunner: map task executor complete.
2026-03-30 14:42:00,030 INFO mapred.LocalJobRunner: Waiting for reduce tasks
```

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Screenshot captured
You can paste the image from the clipbord.

```
                Total committed heap usage (b
        File Input Format Counters
                Bytes Read=74
2026-03-30 14:42:00,026 INFO mapred.LocalJobRunner: Finishing task: attempt_local1368999810_0001_m_000000_0
2026-03-30 14:42:00,027 INFO mapred.LocalJobRunner: map task executor complete.
2026-03-30 14:42:00,030 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2026-03-30 14:42:00,031 INFO mapred.LocalJobRunner: Starting task: attempt_local1368999810_0001_r_000000_0
2026-03-30 14:42:00,038 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-30 14:42:00,038 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, igno
re cleanup failures: false
2026-03-30 14:42:00,038 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-03-30 14:42:00,040 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.reduce.Shuffle@7fe81f49
2026-03-30 14:42:00,042 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-03-30 14:42:00,074 INFO reduce.MergeManagerImpl: MergerManager: memoryLimit=698771008, maxSingleShuffleLimit=174692752, mergeThreshold=461
188896, ioSortFactor=10, memToMemMergeOutputsThreshold=10
2026-03-30 14:42:00,076 INFO reduce.EventFetcher: attempt_local1368999810_0001_r_000000_0 Thread started: EventFetcher for fetching Map Complet
ion Events
2026-03-30 14:42:00,104 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt_local1368999810_0001_m_000000_0 decomp:
 121 len: 125 to MEMORY
2026-03-30 14:42:00,107 INFO reduce.InMemoryMapOutput: Read 121 bytes from map-output for attempt_local1368999810_0001_m_000000_0
2026-03-30 14:42:00,109 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 121, inMemoryMapOutputs.size() -> 1, commitMemor
y -> 0, usedMemory ->121
2026-03-30 14:42:00,111 INFO reduce.EventFetcher: EventFetcher is interrupted.. Returning
2026-03-30 14:42:00,111 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-03-30 14:42:00,112 INFO reduce.MergeManagerImpl: finalMerge called with 1 in-memory map-outputs and 0 on-disk map-outputs
2026-03-30 14:42:00,117 INFO mapred.Merger: Merging 1 sorted segments
2026-03-30 14:42:00,117 INFO mapred.Merger: Down to the last merge-pass, with 1 segments left of total size: 114 bytes
2026-03-30 14:42:00,119 INFO reduce.MergeManagerImpl: Merged 1 segments, 121 bytes to disk to satisfy reduce memory limit
2026-03-30 14:42:00,119 INFO reduce.MergeManagerImpl: Merging 1 files, 125 bytes from disk
2026-03-30 14:42:00,120 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from memory into reduce
2026-03-30 14:42:00,120 INFO mapred.Merger: Merging 1 sorted segments
2026-03-30 14:42:00,121 INFO mapred.Merger: Down to the last merge-pass, with 1 segments left of total size: 114 bytes
2026-03-30 14:42:00,122 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-03-30 14:42:00,406 INFO mapred.Task: Task:attempt_local1368999810_0001_r_000000_0 is done. And is in the process of committing
2026-03-30 14:42:00,413 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-03-30 14:42:00,413 INFO mapred.Task: Task attempt_local1368999810_0001_r_000000_0 is allowed to commit now
2026-03-30 14:42:00,513 INFO output.FileOutputCommitter: Saved output of task 'attempt_local1368999810_0001_r_000000_0' to hdfs://localhost:900
0/r_output
```

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Screenshot captured
You can paste the image from the clipboard.

```
2026-03-30 14:42:00,515 INFO mapred.Task: Fir                                  : Counters: 30
        File System Counters
                FILE: Number of bytes read=3510
                FILE: Number of bytes written=645313
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=74
                HDFS: Number of bytes written=79
                HDFS: Number of read operations=10
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=3
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Combine input records=0
                Combine output records=0
                Reduce input groups=10
                Reduce shuffle bytes=125
                Reduce input records=10
                Reduce output records=10
                Spilled Records=10
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=227540992
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Output Format Counters
                Bytes Written=79
2026-03-30 14:42:00,515 INFO mapred.LocalJobRunner: Finishing task: attempt_local1368999810_0001_r_000000_0
2026-03-30 14:42:00,515 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-03-30 14:42:00,736 INFO mapreduce.Job:  map 100% reduce 100%
```

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hadoop@student: ~/hadoop

```
                Map output bytes=130
                Map output materialized bytes=125
                Input split bytes=91
                Combine input records=14
                Combine output records=10
                Reduce input groups=10
                Reduce shuffle bytes=125
                Reduce input records=10
                Reduce output records=10
                Spilled Records=20
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=12
                Total committed heap usage (bytes)=455081984
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=74
        File Output Format Counters
                Bytes Written=79
hadoop@student:~/hadoop$ hdfs dfs -cat /r_output/part-00000
HDFS    1
Hadoop  2
MapReduce       1
a       2
is      2
of      2
processing      1
storage 1
tool    1
unit    1
hadoop@student:~/hadoop$
```